Magnetic circuits

Answers and worked solutions · Day 03 · Mon 26 Oct · pack M3

5. Practice

From your course files

PQ 3.3 (practice questions).

$\mathcal{R} = \dfrac{l}{\mu_R \mu_0 A}$ each time, with everything in metres.

a. $A = \pi(0.5 \times 10^{-3})^2 = 7.854 \times 10^{-7}$ m², so $\mathcal{R} = \dfrac{0.1}{1000 \times 4\pi \times 10^{-7} \times 7.854 \times 10^{-7}} =$ **101.3 × 10⁶ At/Wb**.

b. $\mathcal{R} = \dfrac{0.2}{5000 \times 4\pi \times 10^{-7} \times 2 \times 10^{-4}} =$ **159.2 × 10³ At/Wb**.

c. $\mathcal{R} = \dfrac{10^{-3}}{4\pi \times 10^{-7} \times 10 \times 10^{-4}} =$ **795.8 × 10³ At/Wb**. A 1 mm gap has five times the reluctance of the whole 20 cm core in (b).

These match the official answers.

PQ 3.4 (practice questions).

$\text{mmf} = Hl = \dfrac{B}{\mu_R \mu_0} l$ and $I = \text{mmf}/N$.

a. $H = 1/(4\pi \times 10^{-7}) = 795.8$ kA/m, so mmf $= 795.8 \times 10^3 \times 10^{-3} =$ **795.8 At** and $I =$ **7.958 A**.

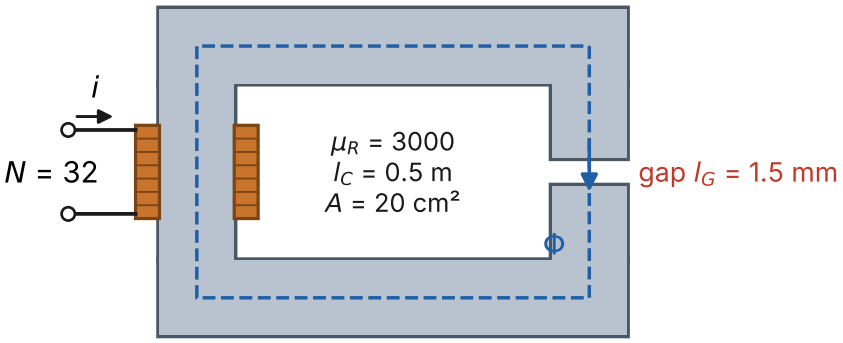
b. $H = 1/(5000 \times 4\pi \times 10^{-7}) = 159.2$ A/m, so mmf $= 159.2 \times 0.2 =$ **31.83 At** and $I =$ **0.3183 A**.

These match the official answers.

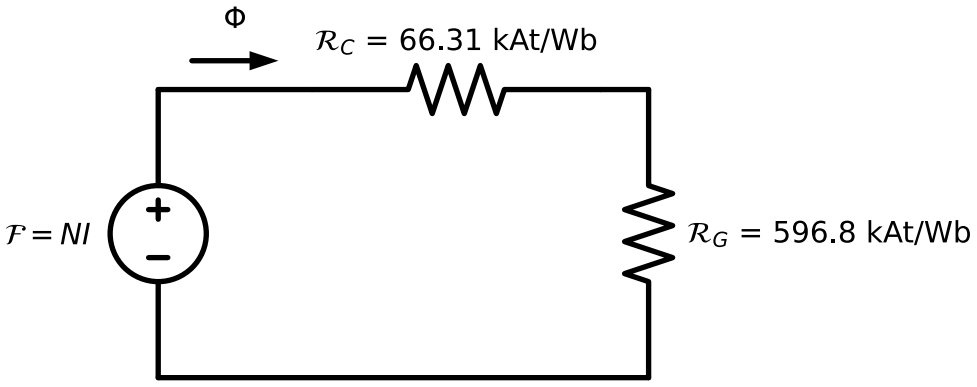
Jun23 Q2a (past exam, 9 marks).

GIVEN: a C-core with $A = 20$ cm² $= 2 \times 10^{-3}$ m², $l_C = 0.5$ m, $\mu_R = 3000$, $N = 32$, and one gap $l_G = 1.5$ mm.

Jun23 Q2a: C-core with one air gap (gap exaggerated)



Jun23 Q2a: C-core with a 32-turn coil and one 1.5 mm gap



Magnetic equivalent circuit for Jun23 Q2a: core and gap reluctances in series

(i) Reluctances and L [4].

$$\mathcal{R}_G = \frac{l_G}{\mu_0 A} = \frac{1.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 2 \times 10^{-3}} = \mathbf{596.8\ kAt/Wb}$$

$$\mathcal{R}_C = \frac{l_C}{\mu_R \mu_0 A} = \frac{0.5}{3000 \times 4\pi \times 10^{-7} \times 2 \times 10^{-3}} = \mathbf{66.31\ kAt/Wb}$$

These are in series: $\mathcal{R}_T = 663.1$ kAt/Wb.

$$L = \frac{N^2}{\mathcal{R}_T} = \frac{32^2}{663.1 \times 10^3} = \mathbf{1.544\ mH}$$

(ii) Maximum energy in the gap only [2]. At $\hat{B} = 1.8$ T, the gap volume is $Al_G = 2 \times 10^{-3} \times 1.5 \times 10^{-3} = 3 \times 10^{-6}$ m³.

$$E = \frac{B^2}{2\mu_0} Al_G = \frac{1.8^2}{2 \times 4\pi \times 10^{-7}} \times 3 \times 10^{-6} = \mathbf{3.867\ J}$$

(For interest: the total $\frac{1}{2}LI^2$ at 1.8 T is 4.30 J. The gap holds 90% of it, matching $\mathcal{R}_G/\mathcal{R}_T = 0.9$.)

(iii) Iron loss at new B and f [3]. Split the 12 W: $P_{H1} = 0.3 \times 12 = 3.6$ W and $P_{E1} = 0.7 \times 12 = 8.4$ W.

$$P_{H2} = 3.6\left(\frac{1.2}{1.5}\right)^2\left(\frac{100}{60}\right) = \mathbf{3.84\ W}$$

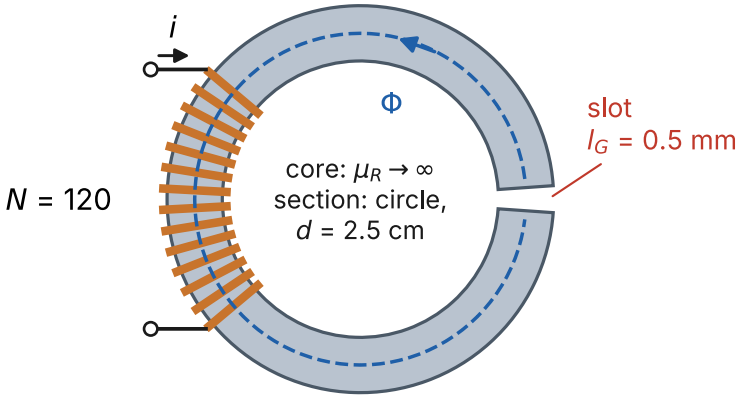
$$P_{E2} = 8.4\left(\frac{1.2}{1.5}\right)^2\left(\frac{100}{60}\right)^2 = \mathbf{14.93\ W}$$

$$P_{FE} = \mathbf{18.77\ W}$$

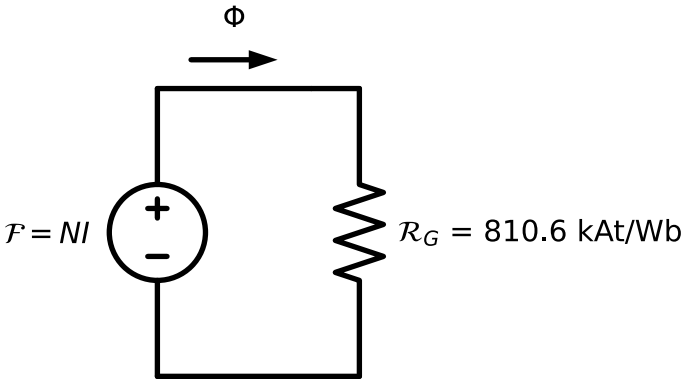
All match the course answers.

Jun24 Q2a (past exam, 7 marks).

GIVEN: a toroid with a circular section of diameter 2.5 cm, so $A = \pi(0.0125)^2 = 4.909 \times 10^{-4}$ m². $N = 120$. The permeability is infinite, so $\mathcal{R}_C = 0$. One slot gives one gap, $l_G = 0.5$ mm.



Jun24 Q2a: toroid of circular section with a 120-turn coil and one 0.5 mm saw-cut slot



$\mathcal{R}_C = 0$ (infinite permeability)

Magnetic equivalent circuit for Jun24 Q2a: with infinite permeability only the gap reluctance remains

(i) [3].

$$\mathcal{R}_G = \frac{0.5 \times 10^{-3}}{4\pi \times 10^{-7} \times 4.909 \times 10^{-4}} = \mathbf{810.6\ kAt/Wb}$$

$$L = \frac{N^2}{\mathcal{R}_G} = \frac{120^2}{810.6 \times 10^3} = \mathbf{17.77\ mH}$$

Note that the mean diameter of the toroid is not given and not needed: with infinite permeability only the gap matters.

(ii) Maximum 50 Hz voltage [2]. Use the back-emf equation. It needs no reluctance.

$$V_{\max} = 4.44\, N f \hat{B}_{\max} A = 4.44 \times 120 \times 50 \times 1.5 \times 4.909 \times 10^{-4} = \mathbf{19.62\ V\ rms}$$

(iii) Force across the slot [2]. There is one gap, so the area is used once.

$$F = \frac{B^2}{2\mu_0} A = \frac{1.5^2}{2 \times 4\pi \times 10^{-7}} \times 4.909 \times 10^{-4} = \mathbf{439.5\ N}$$

All match the course answers.

2023 quiz Q3 (6 marks).

GIVEN: a toroid with mean diameter 80 mm, so $l = \pi(0.08) = 0.2513$ m. $A = 20$ cm² $= 2 \times 10^{-3}$ m², $\mu_R = 1000$, $\hat{B} = 1.5$ T.

(a) Turns for 240 V, 50 Hz [2].

$$N = \frac{240}{4.44 \times 50 \times 1.5 \times 2 \times 10^{-3}} = 360.4 \ \Rightarrow\ \mathbf{N = 360\ (or\ 361)}$$

Rounding up to 361 keeps $\hat{B} \le 1.5$ T. The answer sheet accepts both.

(b) L and X for N = 200 [3].

$$\mathcal{R} = \frac{0.2513}{1000 \times 4\pi \times 10^{-7} \times 2 \times 10^{-3}} = 100.0\ \text{kAt/Wb}$$

$$L = \frac{200^2}{100 \times 10^3} = \mathbf{0.400\ H}, \qquad X = 2\pi(50)(0.4) = \mathbf{125.7\ \Omega}$$

(c) Current at 100 V [1]. $I = V/X = 100/125.66 = \mathbf{0.796\ A}$. The course gives 0.7955 A because it used X rounded to 125.6 Ω. Unrounded, the answer is 0.7958 A.

2023 quiz Q4 (4 marks).

GIVEN: two poles with one 4 mm gap between them. The pole area is 20 cm², $N = 75$, $B = 1.1$ T, and the core is infinitely permeable.

(a) [2]. All the mmf is across the gap: $NI = H_G l_G = (B/\mu_0) l_G$.

$$I = \frac{B l_G}{\mu_0 N} = \frac{1.1 \times 0.004}{4\pi \times 10^{-7} \times 75} = \mathbf{46.69\ A}$$

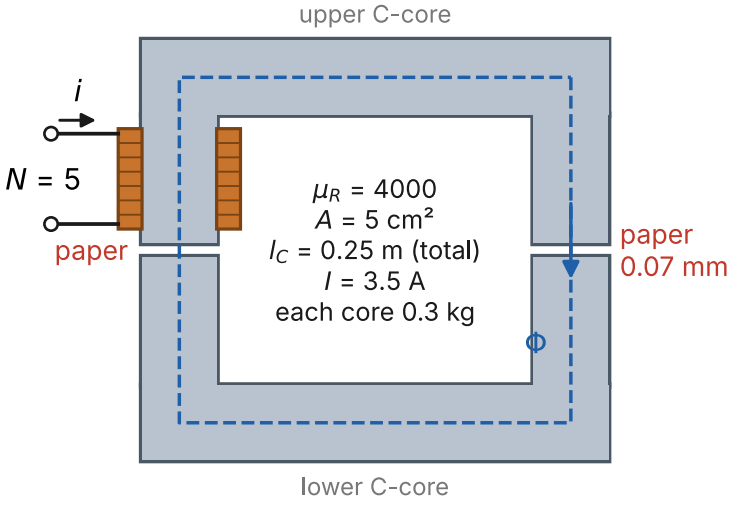
(b) [2]. There is one gap, so one pole-face pair:

$$F = \frac{1.1^2}{2\mu_0} \times 20 \times 10^{-4} = \mathbf{962.9\ N}$$

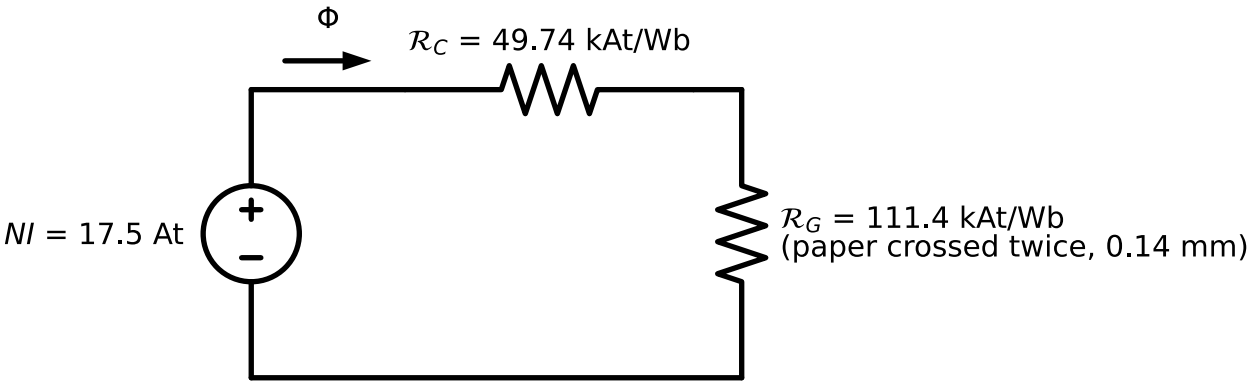
2024 test Q4 (7 marks).

GIVEN: two identical C-cores. $A = 5$ cm² $= 5 \times 10^{-4}$ m², $\mu_R = 4000$, the total iron path is $l_C = 0.25$ m, $N = 5$, $I = 3.5$ A (so $NI = 17.5$ At), and each core has $m = 0.3$ kg.

2024 test Q4: two C-cores with paper between them



2024 test Q4: two C-cores with a 5-turn coil, separated by 0.07 mm paper at both joints



Magnetic equivalent circuit for 2024 test Q4(b): iron and paper-gap reluctances in series

(a) No gap [2]. $NI = H_C l_C = B l_C/(\mu_R \mu_0)$, so

$$B = \frac{\mu_R \mu_0 NI}{l_C} = \frac{4000 \times 4\pi \times 10^{-7} \times 17.5}{0.25} = \mathbf{0.3519\ T}$$

(b) 0.07 mm paper [3]. The flux crosses the paper twice per loop, so $l_G = 0.14$ mm.

$$NI = B\left(\frac{l_C}{\mu_R \mu_0} + \frac{l_G}{\mu_0}\right) = B(49.74 + 111.4) \times 10^3$$

$$B = \frac{17.5}{161.1 \times 10^3} = \mathbf{0.1086\ T}$$

A sheet of paper has more than twice the reluctance of the whole iron path.

(c) Minimum B to hold the lower core [2]. The weight is $mg = 0.3 \times 9.81 = 2.943$ N. Both pole faces pull, so the force area is $2A = 10^{-3}$ m².

$$B = \sqrt{\frac{2\mu_0 mg}{2A}} = \sqrt{\frac{2 \times 4\pi \times 10^{-7} \times 2.943}{10^{-3}}} = \mathbf{0.0860\ T}$$

This is below 0.1086 T, so the magnet still holds with the paper in place.

2024 test Q5 (5 marks).

GIVEN: a 240 V/48 V, 50 Hz transformer with $A = 40$ cm² $= 4 \times 10^{-3}$ m² and $\hat{B}_{\max} = 1.6$ T. The path length and μ_R are not needed here.

(a) [2].

$$N_1 = \frac{240}{4.44 \times 50 \times 1.6 \times 4 \times 10^{-3}} = 168.9 \ \Rightarrow\ \mathbf{169\ turns}$$

This gives $\hat{B} = 1.599$ T, which is just under the limit.

(b) [2]. $A_w = \pi(0.25 \times 10^{-3})^2 = 1.963 \times 10^{-7}$ m².

$$R = \frac{l}{\sigma A_w} = \frac{40}{59 \times 10^6 \times 1.963 \times 10^{-7}} = \mathbf{3.453\ \Omega}$$

(c) [1]. $N_2 = 169 \times 48/240 = 33.8 \ \Rightarrow\ \mathbf{34\ turns}$.

2023 quiz Q7 (4 marks).

- **Process.** The back-emf equation links the primary voltage, frequency, turns and peak core flux. For a given core area and supply, choose the number of primary turns so that the peak flux density sits just below the saturation level of the core material.
- **Too few turns.** The peak flux density must be higher. The core saturates and the magnetising current becomes very large and distorted. The winding overheats.
- **Too many turns.** The core runs at a low flux density, so it is under-used. The longer wire has more resistance, which raises copper loss and lowers efficiency, and can also cause overheating.

(Exam instruction: words only, so no equations or sketches.)

PQ 3.10 (practice questions).

Hint: $\mathcal{R}_C = 0$; $l_G = \mu_0 N^2 A/L$; $I_{\max} = N\Phi_{\text{sat}}/L$.

a. Only the gap has reluctance, so $L = N^2/\mathcal{R}_G = \mu_0 N^2 A/l_G$:

$$l_G = \frac{\mu_0 N^2 A}{L} = \frac{4\pi \times 10^{-7} \times 50^2 \times 5 \times 10^{-4}}{10^{-3}} = \mathbf{1.571\ mm}$$

b. $\mathcal{R}_G = 10^{-3}/(4\pi \times 10^{-7} \times 5 \times 10^{-4}) = 1.592 \times 10^6$ At/Wb, so $N = \sqrt{L\mathcal{R}_G} = \sqrt{1592} = 39.89$, which rounds to **40 turns**.

c. $\Phi_{sat} = B_{sat}A = 0.3 \times 5 \times 10^{-4} = 1.5 \times 10^{-4}$ Wb. From $N\Phi = LI$: $I_{\max} = 50 \times 1.5 \times 10^{-4}/10^{-3} =$ **7.5 A**.

These match the official answers.

PQ 3.13 (practice questions).

Hint for (c): the loss factor at 400 Hz is $(8 + 64)/2 = 36$ at fixed B, so $\hat{B} = 1/\sqrt{36}$.

a. Both loss types go as $\hat{B}^2$: $3 \times 1.5^2 =$ **6.75 W**.

b. The frequency rises ×8. (i) $P_H \propto f$: $3 \times 8 =$ **24 W**. (ii) $P_E \propto f^2$: $3 \times 64 =$ **192 W**. (iii) 1.5 W of each: $1.5 \times 8 + 1.5 \times 64 = 12 + 96 =$ **108 W**.

c. Need $108\,\hat{B}^2 = 3$, so $\hat{B} = \sqrt{3/108} = 1/6 =$ **0.1667 T**.

These match the official answers.

PQ 3.14 (practice questions).

a. In the **airgap** (it has most of the reluctance, so most of the mmf and energy).

b. **No**, because air is not conductive (no eddy currents) and has no magnetic domains (no hysteresis).

c. **Eddy-current** loss and **hysteresis** loss.

d. **Eddy-current** loss, because it is proportional to frequency squared, so it dominates at high frequencies.

e. To **reduce eddy-current losses**: the insulated laminations break up the eddy-current paths.

f. It **increases rapidly and becomes distorted** (peaky), because μ_R collapses and much more current is needed for each extra bit of flux.

These match the official answers.

New questions

N1. Gapped rectangular core [5 marks].

Magnetic equivalent circuit for New question 1

$A = 1.2 \times 10^{-3}$ m².

a. $\mathcal{R}_C = \dfrac{0.6}{2500\mu_0 \times 1.2 \times 10^{-3}} = 159.2\ \text{kAt/Wb}, \qquad \mathcal{R}_G = \dfrac{0.8 \times 10^{-3}}{\mu_0 \times 1.2 \times 10^{-3}} = 530.5\ \text{kAt/Wb}$

$\mathcal{R}_T = 689.7$ kAt/Wb, so $L = 250^2/689.7 \times 10^3 = \mathbf{90.6\ mH}$. Without the gap it would be 393 mH.

b. $\Phi = 1.2 \times 1.2 \times 10^{-3} = 1.44$ mWb. Then $NI = \Phi\mathcal{R}_T = 993.1$ At, so $I = \mathbf{3.97\ A}$.

The gap takes $\mathcal{R}_G/\mathcal{R}_T = \mathbf{76.9\%}$ of the mmf (764 At of 993 At).

N2. Shell (E) core with parallel paths [7 marks].

Magnetic equivalent circuit for New question 2: centre-limb and gap reluctances in series with the two outer-path reluctances in parallel

a. The circuit is: source NI in series with $\mathcal{R}_{\text{centre}}$, $\mathcal{R}_G$, and the two outer reluctances in parallel.

$$\mathcal{R}_{\text{centre}} = \frac{0.10}{2000\mu_0(8 \times 10^{-4})} = 49.74\ \text{k}, \quad \mathcal{R}_G = \frac{10^{-3}}{\mu_0(8 \times 10^{-4})} = 994.7\ \text{k}, \quad \mathcal{R}_{\text{outer}} = \frac{0.30}{2000\mu_0(4 \times 10^{-4})} = 298.4\ \text{k}$$

All are in At/Wb. The two equal outer paths in parallel give $\mathcal{R}_{\parallel} = 298.4/2 = 149.2$ kAt/Wb.

$$\mathcal{R}_T = 49.74 + 994.7 + 149.2 = \mathbf{1194\ kAt/Wb}$$

b. $L = 200^2/1.194 \times 10^6 = \mathbf{33.5\ mH}$.

c. $\Phi = 1.2 \times 8 \times 10^{-4} = 0.96$ mWb. Then $NI = \Phi\mathcal{R}_T = 1146$ At, so $I = \mathbf{5.73\ A}$.

The flux splits equally: 0.48 mWb in each outer limb. So $B_{\text{outer}} = 0.48 \times 10^{-3} / 4 \times 10^{-4} = \mathbf{1.2\ T}$. Shell cores make the outer limbs half the centre area for exactly this reason: the flux density stays uniform.

N3. Back-emf design and loss scaling [8 marks].

a. $N = 230/(4.44 \times 50 \times 1.5 \times 1.5 \times 10^{-3}) = 460.5$, so round up to **461 turns**. Then $\hat{B} = 230/(4.44 \times 461 \times 50 \times 1.5 \times 10^{-3}) = \mathbf{1.498\ T}$.

b. $\hat{B} \propto 1/f$ at fixed V, so $\hat{B} = 1.498 \times 50/60 = \mathbf{1.249\ T}$.

The base split is $P_{H1} = 8$ W and $P_{E1} = 12$ W.

- $P_H = 8 \times (5/6)^2 \times (6/5) = \mathbf{6.67\ W}$ (this is the $\propto 1/f$ case).
- $P_E = 12 \times (5/6)^2 \times (6/5)^2 = \mathbf{12.0\ W}$ (unchanged).
- Total **18.67 W**.

c. At 40 Hz and 230 V, $\hat{B} = 1.498 \times 50/40 = 1.87$ T. That is past the knee, so the core saturates and the magnetising current becomes large and distorted, overheating the winding.

$V_{\max} = 4.44 \times 461 \times 40 \times 1.5 \times 1.5 \times 10^{-3} = \mathbf{184.2\ V}$. This is the constant-V/f rule: $230 \times 40/50 = 184$ V.

At $\hat{B} = 1.5$ T, 40 Hz: $P_H = 8 \times (1.5/1.498)^2 \times 0.8 = 6.42$ W and $P_E = 12 \times (1.5/1.498)^2 \times 0.64 = 7.70$ W, for a total of **14.1 W**. Treating $\hat{B}$ as unchanged gives 14.08 W, which is also acceptable.

N4. Electromagnetic lifter, two gaps [6 marks].

Magnetic equivalent circuit for New question 4: two gap reluctances in series, iron reluctance zero

a. There are two pole faces, so the force area is $2A = 1.2 \times 10^{-3}$ m².

$$B = \sqrt{\frac{2\mu_0 F}{2A}} = \sqrt{\frac{2\mu_0 (150)}{1.2 \times 10^{-3}}} = 0.5605\ \text{T}$$

The total gap is $l_G = 3$ mm, so

$$I = \frac{B l_G}{\mu_0 N} = \frac{0.5605 \times 0.003}{4\pi \times 10^{-7} \times 400} = \mathbf{3.35\ A}$$

b. $\mathcal{R}_G = 0.003/(\mu_0 \times 6 \times 10^{-4}) = 3.979$ MAt/Wb, so $L = 400^2/3.979 \times 10^6 = \mathbf{40.2\ mH}$.

$E = \frac{B^2}{2\mu_0} A l_G = \mathbf{0.225\ J}$. Check: $\frac{1}{2} L I^2 = 0.225$ J.

c. The gap falls by a factor of 3 at the same NI, so B rises ×3 to 1.68 T and F rises ×9 to about 1350 N. But 1.68 T is at or beyond the knee of the B–H curve. The iron saturates, so the real force will be lower than 1350 N and the "infinite permeability" assumption fails.

N5. Gapped inductor design, chained [8 marks].

a. $N\hat{B}A = L\hat{I}$, so $N = L\hat{I}/(\hat{B}A) = (5 \times 10^{-3} \times 10)/(1.2 \times 6 \times 10^{-4}) = 69.4$. Round up to **70 turns**.

b. $L = N^2 \mu_0 A / l_G$, so

$$l_G = \frac{\mu_0 N^2 A}{L} = \frac{4\pi \times 10^{-7} \times 70^2 \times 6 \times 10^{-4}}{5 \times 10^{-3}} = \mathbf{0.739\ mm}$$

c. $\hat{B} = \mu_0 N \hat{I} / l_G = \mathbf{1.190\ T}$, which is at or below 1.2 T.

$\frac{1}{2} L \hat{I}^2 = \frac{1}{2}(5 \times 10^{-3})(100) = 0.250$ J, and $\frac{\hat{B}^2}{2\mu_0} A l_G = 0.250$ J. They are equal because all the energy is in the gap when $\mu_R \to \infty$.

d. $E = 4.44 \times 70 \times 50 \times 1.190 \times 6 \times 10^{-4} = \mathbf{11.10\ V\ rms}$.

Check: $\omega L I_{\text{rms}} = 2\pi(50)(5 \times 10^{-3})(10/\sqrt{2}) = 11.11$ V. The tiny difference comes from rounding $\sqrt{2}\pi$ to 4.44.

6. Quick recall check

Question

1. At/Wb (A/Wb, or H^{-1}). mmf NI is the analogue of emf (voltage).
2. L roughly halves. B roughly halves (since $\Phi = NI/\mathcal{R}$). The force falls to about a quarter ($F \propto B^2$).
3. ×0.25. Both P_H and P_E go as $\hat{B}^2 \propto V^2$ (Jun24 Q2c(iii)).
4. $\hat{B} \times 5/6$. $P_H \propto \hat{B}^2 f$ gives $\times 5/6$. $P_E \propto \hat{B}^2 f^2$ gives ×1 (unchanged).
5. 120 V. $\hat{B} \propto V/f$, so the full 240 V at 25 Hz would double $\hat{B}$ and saturate the core.